

Project Initialization and Planning Phase

Date	08 October 2026
Team ID	UNM1611Firefox
Project Title	AI StudyBuddy API
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution is an AI-powered study assistant backend designed to help students manage study materials and generate useful learning resources. The system uses a RESTful API with Google Gemini AI to transform uploaded study materials into summaries, flashcards, quizzes, and personalized study plans, making exam preparation more organized and efficient.

Project Overview	
Objective	The primary objective is to develop an AI-powered study assistant API that helps students manage study materials and generate personalized learning resources such as summaries, flashcards, quizzes, and study plans.
Scope	The project covers secure user authentication, study-material upload and management, database storage, and AI-powered generation of summaries, flashcards, quizzes, and personalized study plans using Google Gemini AI.
Problem Statement	
Description	Students often deal with lengthy and scattered study materials and spend significant time manually preparing summaries, revision notes, questions, and study plans. They need a simple and integrated solution that can organize learning materials and provide AI-powered study assistance.
Impact	The lack of an integrated study assistant can make exam preparation time-consuming, stressful, and difficult to organize. AI StudyBuddy aims to reduce manual effort and make learning and revision more efficient.
Proposed Solution	
Approach	Develop a secure RESTful backend using Node.js and Express.js, with MongoDB for data storage and Google Gemini AI for processing study materials and generating useful learning resources.
Key Features	- Secure user registration and login using JWT authentication - Study-material upload and management - AI-generated summaries - AI-generated flashcards - AI-generated quizzes - Personalized study-plan generation - MongoDB-based storage - RESTful API architecture

	- Modular backend architecture - API testing using Postman - Secure environment-variable management for API keys - Easy integration with a frontend application
--	--

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computer / Laptop for development and testing	Minimum 8 GB RAM, modern multi-core processor
Memory	RAM for backend development, database and testing	8 GB RAM or higher
Storage	Storage for source code, database and study materials	Minimum 10 GB available storage
Software		
Frameworks	Backend development frameworks	Node.js, Express.js
Libraries	Required backend and AI integration libraries	- Mongoose - dotenv - bcryptjs - jsonwebtoken - @google/generative-ai
Development Environment	IDE / API testing / database tools	- Visual Studio Code - Postman - MongoDB Compass
Data		
Data	Study materials and user-related data	- User account information - Uploaded study materials - Generated summaries - Flashcards - Quizzes - Study plans
AI / External Service		
AI Service	Generative AI processing	Google Gemini API